

Jinil Kim

haba6030@snu.ac.kr | Homepage: <https://haba6030.github.io/> | GitHub: github.com/haba6030

RESEARCH INTEREST

- Developing computational frameworks that extract latent signatures from human brain and behavior
- Translating these into real-world systems for personalized interventions and human-inspired AI
- Uncovering shared and distinct principles across brains, machines, and languages

EDUCATION

Seoul National University

B.A. Linguistics · B.A. Psychology · B.S. Computer Science

Seoul, South Korea

Mar 2021 – Exp. Feb 2027

GPA: 4.18 / 4.3 (3.96 / 4.0) | Major GPA (4.0): Ling./Psych.: 4.00, CS: 3.82

* Mandatory military service, Jul 2023 – Jan 2025

The University of Hong Kong

Summer Research Program, Faculty of Psychology

Hong Kong SAR

Jun 2025 – Aug 2025

PUBLICATIONS & MANUSCRIPTS

Sole first author on all listed work.

- **Kim, J.**, Cho, A. M., Seo, J., & Cha, J. (2026). *Inferring Individualized Color-Vision Distortions from fMRI Hue-Representation Geometry*. ICML 2026 Workshop on Structured Data for Health (SD4H), Seoul, Korea. [Workshop paper — accepted]
- **Kim, J.**, Cho, M., Seo, J., & Cha, J. (2026). *fMRI Decoding Reveals Intact Neural Color Representations in Color Vision Deficiency*. OHBM 2026 Annual Meeting, Bordeaux, France. [Poster — accepted]
- **Kim, J.**, & Yao, Y.-W. *When, not just how strongly: a delayed onset of attribute integration in depression*. Manuscript in preparation. (Earlier version presented as poster, HKU Summer Research Program, Aug 2025.)
- **Kim, J.**, et al. Manuscript in preparation on inverse reinforcement learning of human planning.

RESEARCH PROJECTS

Distortion of cortical hue geometry in color vision deficiency enables individualized color-correction filters

Project Lead, sole first author (Advisor: Prof. Jiook Cha)

Apr 2025 – Present

- Characterized individual-specific distortions of cortical hue geometry in color vision deficiency from fMRI, and translated them into personalized color-correction filters.
- Funded by the SNU Undergraduate Research Grant (\$3,500).

When, not just how strongly: a delayed onset of attribute integration in depression

Project Lead, sole first author (Advisor: Prof. Yuan-Wei Yao)

Jun 2025 – Present

- Built a time-varying drift-diffusion model with an onset-latency parameter (τ) showing that depression integrates the second decision attribute later in social decisions.

Inverse Reinforcement Learning of Latent Human Planning

Project Lead, sole first author (Advisor: Prof. Woo-Young Ahn)

Sep 2025 – Present

- Developing an inverse reinforcement learning framework that more faithfully models human decision-making while recovering latent planning depth from behavior alone.

RESEARCH EXPERIENCE

Connectome Lab Undergraduate RA (Advisor: Prof. Jiook Cha) Research area: Brain foundation models; neural fields	Seoul National University Jul 2026 – Present
Computational Clinical Science Lab Undergraduate RA (Advisor: Prof. Woo-Young Ahn) Research area: Inverse RL; computational psychiatry; multimodal digital phenotyping	Seoul National University Apr 2025 – Present
Mental health, Internet, Neuroscience & Decision-making (MIND) Lab Visiting Researcher (Advisor: Prof. Yuan-Wei Yao) Research area: Computational modeling of social decision-making	University of Hong Kong Jun 2025 – Aug 2025

GRANTS

KOSAF Humanity 100 Years Undergraduate Scholarship Amount: \$12,500 (KRW 17,000,000)	Mar 2025 – Jul 2026
SNU Undergraduate Research Grant (color-vision project) Amount: \$3,500	Apr 2025
The University of Hong Kong Foundation for Educational Development and Research Scholarship Amount: \$1,300 (HKD 10,000)	Jun 2025 – Aug 2025
Superior Academic Performance scholarship	Mar 2022, 2023; Sep 2022

HONORS & AWARDS

Best Poster Award in SNU Undergrad Research Program (4th Place)	Jan 2026
SNU Liberal Arts Competition (1st Place)	Jul 2023

TEACHING

Teaching Assistant, Basic Computing English (Mar 2026) · Korean (Sep 2025)	Seoul National University
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LANGUAGES

Human: Korean (Native), English (Full Professional Proficiency), French (Beginner)
Programming: Python, R, Stan, C++, Java